Monocular Visual-Inertial Odometry: Implementation and Evaluation on TUM VI

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Introduction

Real-time state estimation is a core challenge in robotics navigating GPS-denied environments.

Monocular Visual Odometry (VO) tracks camera trajectories using spatial features but suffers from:

- Unobservable absolute metric scale.
- Cumulative tracking drift over extended trajectories.

Visual-Inertial Odometry (VIO) tightly couples discrete visual observations (20 Hz) with continuous inertial measurements (200 Hz). This fusion renders the metric scale observable and bridges tracking gaps caused by motion blur or textureless surfaces.

System Pipeline Architecture

Our system from-scratch implementation is modeled after the VINS-Mono framework. The pipeline processes asynchronous sensor streams, recovers metric scale during initialization, and stabilizes within a tightly-coupled sliding-window optimizer.

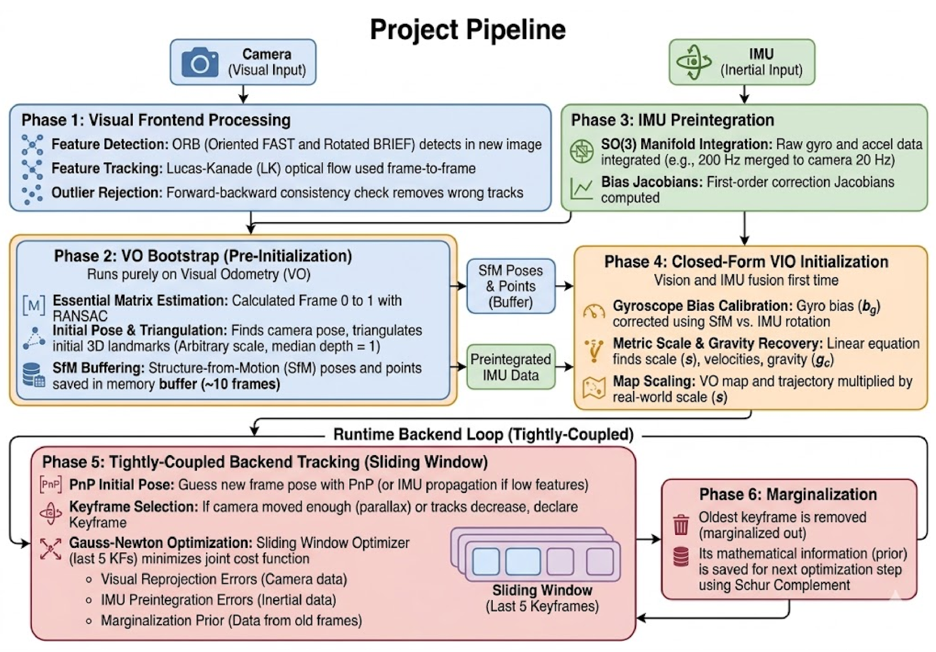

Fig. 1: Complete VIO System Pipeline

Visual Frontend

- **Detection:** ORB (Oriented FAST and Rotated BRIEF) features.
- **Tracking:** Lucas-Kanade (LK) optical flow.
- **Outlier Rejection:** Forward-backward consistency check (> 1.0 pixel error rejected).

Visual Reprojection Residual ($r_{\mathcal{V}}$):

$$r_{\mathcal{V}}(\hat{z}_l^m, x_l, \mathcal{P}_m) = \begin{bmatrix} \hat{u}_l \\ \hat{v}_l \end{bmatrix} - \pi(R_w^{c_l}(\mathcal{P}_m^w - p_w^{c_l}))$$

VO Bootstrap Phase

Before fusing IMU data, a pure visual bootstrap phase initializes the local structure:

- **Essential Matrix:** Computed between frames 0 and 1 using a RANSAC-backed 8-point algorithm.
- **Scale Normalization:** Triangulated 3D landmark point cloud is normalized so the median depth equals exactly 1.0 unit.
- **SfM Buffering:** Buffers ~10 frames to align with inertial vectors.

IMU Preintegration & Initialization

IMU Preintegration on SO(3): Raw IMU measurements ($\tilde{\omega}$, $\tilde{a}$) are accumulated between keyframes i and j to avoid reintegration during optimization:

$$\Delta R_{ij} = \prod_{k=i}^{j-1} \mathrm{Exp}((\tilde{\omega}_k - b_{g_i})\Delta t)$$

$$\Delta v_{ij} = \sum_{k=i}^{j-1} \Delta R_{ik}(\tilde{a}_k - b_{a_i})\Delta t$$

$$\Delta p_{ij} = \sum_{k=i}^{j-1} \left[\Delta v_{ik}\Delta t + \frac{1}{2}\Delta R_{ik}(\tilde{a}_k - b_{a_i})\Delta t^2\right]$$

Closed-Form Initialization:

1. Gyroscope biases (b_g) are calibrated by aligning SfM and IMU rotations across buffered frames.
2. A joint linear system $H_{init}X_{init} = J_{init}$ solves simultaneously for velocities (v_b), gravity (g_c), and the metric scale factor (s).

Sliding-Window Optimization

The core backend minimizes a joint non-linear cost function over a bounded sequence of $N = 5$ active keyframes:

$$\min_{\mathcal{X}} \left\{ \|r_p\|^2_{\Sigma_p} + \sum_{k\in\mathcal{B}} \|r_{\mathcal{I}}\|^2_{\Sigma_{\mathcal{I}}} + \sum_{(l,m)\in\mathcal{C}} \|r_{\mathcal{V}}\|^2_{\Sigma_{\mathcal{V}}} \right\}$$

where $r_{\mathcal{V}}$ is the visual reprojection error, $r_{\mathcal{I}}$ is the IMU preintegration error, and r_p is the marginalization prior.

Schur Complement Marginalization:
To maintain a bounded processing time, the oldest keyframe is removed from the active window when full. Its connected states are eliminated using the Schur complement, transforming its constraints into a dense analytical linear prior (r_p) for the next optimization step.

Experimental Setup

Evaluated on the highly dynamic **TUM VI Benchmark**:

- *room2*: Indoor, structured, dense ground-truth.
- *corridor3*: Long corridor, repetitive geometry.
- *outdoors5*: Expansive outdoor, sudden illumination changes.

Metrics: Absolute Trajectory Error (ATE) and Relative Pose Error (RPE). Evaluated using $SE(3)$ rigid body alignment.

Results: Tracking Performance

The baseline VO tracks motion effectively in localized environments (e.g., room2). However, in highly dynamic and expansive sequences, the tightly-coupled VIO pipeline experiences catastrophic quadratic drift.

Table 1: Trajectory Error Metrics

Sequence	Method	Mean ATE (m)	End Drift (m)
room2	Monocular VO	1.203	N/A
	Full VIO	42.489	N/A
corridor3	Monocular VO	0.880	0.226
	Full VIO	300.159	252.895
outdoors5	Monocular VO	256.977	223.127
	Full VIO	1.64×10^7	8.46×10^6

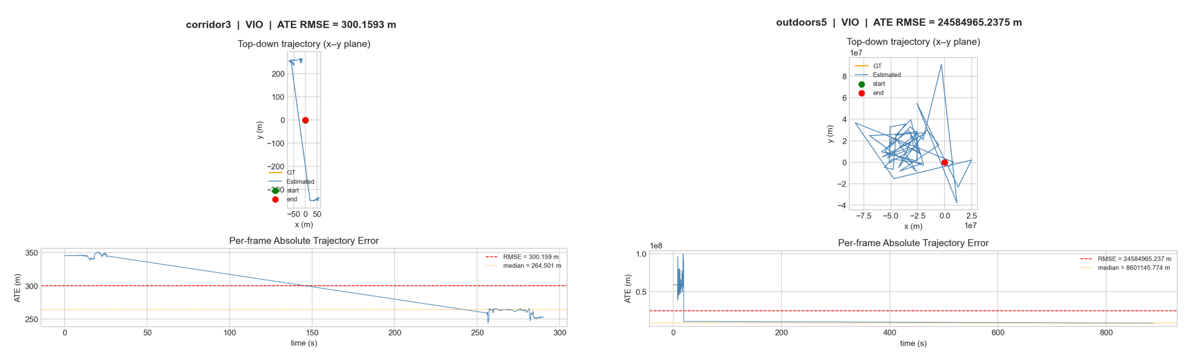

Fig. 2: Left: Catastrophic Drift (corridor3). Right: Catastrophic Drift (outdoors5).

Failure Case & Ablation Study

Information Matrix Imbalance:
The tightly-coupled VIO suffers from numerical instability when solving the Gauss-Newton normal equations ($H\Delta x = b$). The Hessian H is constructed as:

$$H = J_{\mathcal{V}}^T\Sigma_{\mathcal{V}}^{-1}J_{\mathcal{V}} + J_{\mathcal{I}}^T\Sigma_{\mathcal{I}}^{-1}J_{\mathcal{I}} + H_{prior}$$

The IMU information matrix ($\Sigma_{\mathcal{I}}^{-1} \approx 10^9$) outweighs visual constraints ($\Sigma_{\mathcal{V}}^{-1} \approx 10^5$) by several orders of magnitude. The solver treats the IMU as absolute ground truth and assigns impossible velocities to minimize the mathematical cost function.

Impact of Online Bias Estimation:
Disabling online bias estimation ($\Delta b_g = 0, \Delta b_a = 0$) accelerates trajectory divergence, confirming its structural necessity.

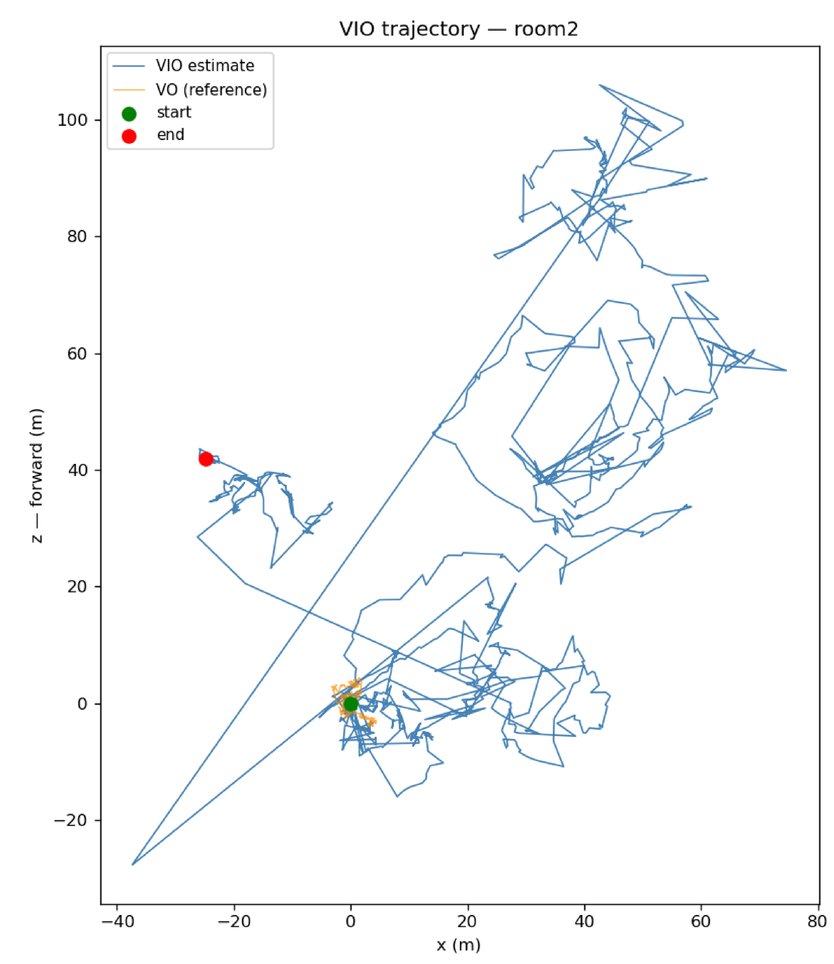

Fig. 3: Ablation: Frozen Biases (room2)

Conclusion

- Developed a tightly-coupled sliding-window VIO system from scratch.
- Highlighted severe challenges in visual-inertial information balancing within the Gauss-Newton optimizer.
- **Future Work:** Implement adaptive covariance scaling to down-weight IMU information and integrate global pose-graph optimization.